

Remainder theorem



- 1
- | | | | | | | | |
|---|-------------------|---|-------------------|---|--------------------|---|--------------------|
| a | $x = -3$ | b | $x = 8$ | c | $x = 6$ | d | $x = -10$ |
| e | $x = \frac{7}{6}$ | f | $x = \frac{1}{3}$ | g | $x = -\frac{1}{2}$ | h | $x = -\frac{5}{4}$ |
- 2
- | | | | | | | | |
|---|-------|---|------------------|---|------------------|---|------------------|
| a | -93 | b | 2100 | c | $\frac{125}{27}$ | d | -5 |
| e | 67 | f | $\frac{2205}{8}$ | g | $-\frac{41}{4}$ | h | $\frac{428}{81}$ |
- 3
- | | | | | | | | |
|---|----------|---|---------|---|---------|---|----------|
| a | $k = -4$ | b | $k = 3$ | c | $k = 4$ | d | $k = -4$ |
| e | $k = -1$ | f | $k = 2$ | g | $k = 3$ | h | $k = -1$ |
- 4 $n = 6$
- 5
- a 0
- b In part (a), we found that $x^{99} + 1$ is divisible by $x + 1$. That is, $x^{99} + 1 = (x + 1) \times Q(x)$ (Equation 1) for some quotient $Q(x)$.
Substituting $x = 4$ into Equation 1, we get $4^{99} + 1 = 5 \times Q(4)$. Therefore, $4^{99} + 1$ is divisible by 5.

Factor theorem

- 6 Factoring
- 7 Expanding or multiplying
- 8 When it is written as a product of prime polynomials.
- 9 The factor theorem states that if f is a polynomial function and $f(c) = 0$, then $(x - c)$ is a factor of $f(x)$.
- 10 Using the factor theorem, $P(-2) = (-2)^4 + 3(-2)^3 - 7(-2)^2 - 27(-2) - 18 = 0$.



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a Using the factor theorem, $P(-4) = 2(-4) + 3(-4)^2 + (-4) + 4 = 0$.

b $(2x^2 + 1)$

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a Using the factor theorem, $P(5) = 10 - 17(5) + 8(5)^2 - (5)^3 = 0$.

b $(x^2 - 3x + 2)$

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a $p = -3$

b $k = -84$

14 $Q(x) = x^3 - 4x^2 - 20x + 3$ because dividing $Q(x)$ by $(x + 3)$ gives a remainder of 0 and $Q(x)$ is the only polynomial such that $Q(-3) = 0$.

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a $(x - 1)(x - 2)(x + 5)$

b $(x - 5)(x + 2)(x - 3)$

c $(x - 3)(x - 5)(x - 1)$

d $(x + 2)(x + 1)(x + 5)$

e $(2x - 5)(x - 1)(x - 2)$

f $(5x - 1)(x - 5)(x - 3)$

g $(2x + 3)(x - 2)(x + 1)$

h $(4x + 3)(x - 2)(x + 1)$